



WORKER HEALTH AND SAFETY MONITORING SYSTEM FOR CONSTRUCTION INDUSTRIES

EXERCISE NO – 09

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9. Worker health and safety monitoring system for construction industries

1. Project overview:

- The Worker Health and Safety Monitoring System is an IoT-based system designed to continuously monitor the health and environmental safety conditions of construction workers. The system uses an ESP8266 NodeMCU as the main controller and collects data from a Pulse Sensor, DHT11 temperature/humidity sensor, MQ2 gas sensor, and NEO-6M GPS module.
- The Pulse Sensor monitors the worker's heart rate, while the DHT11 measures the surrounding temperature and humidity. The MQ2 sensor detects the presence of smoke or combustible gases and generates a gas-alert condition. The NEO-6M GPS provides the worker's latitude, longitude, and movement information for location monitoring.
- The ESP8266 processes all sensor readings and connects to the Internet through Wi-Fi. The collected data is uploaded to ThingSpeak Cloud, where the parameters can be monitored through graphs and numerical displays. If unsafe conditions such as excessive temperature, gas detection, or abnormal heart-rate conditions are detected, the system generates an unsafe status.

2. Problem Statement:

Construction workers are exposed to various hazards such as extreme temperatures, harmful gases, smoke, and physically demanding working conditions.

Traditional safety monitoring methods may not provide continuous information about a worker's health condition or location during an emergency.

There is a need for an IoT-based real-time monitoring system that can monitor important worker health and environmental parameters and identify unsafe conditions.

The proposed system uses an ESP8266, Pulse Sensor, DHT11, MQ2 gas sensor, and NEO-6M GPS to collect safety-related data and transmit it to ThingSpeak Cloud for remote monitoring and analysis.

The system helps provide early warnings when abnormal conditions are detected, thereby supporting faster safety responses and improving overall worker protection.

3. Proposed Solution:

- The proposed solution is an IoT-based Worker Health and Safety Monitoring System designed to continuously monitor construction workers' health, environmental conditions, and location in real time.
- The system uses an ESP8266 NodeMCU as the central controller. A Pulse Sensor monitors the worker's heart rate, while the DHT11 sensor measures temperature and humidity. An MQ2 gas sensor detects the presence of smoke or combustible gases in the working environment. A NEO-6M GPS module tracks the worker's geographical location using latitude and longitude.
- The ESP8266 collects and processes the sensor readings and connects to the Internet through Wi-Fi. The collected data is uploaded to ThingSpeak Cloud, where supervisors can remotely monitor the worker's health and safety parameters through a dashboard.
- When the system detects conditions such as gas detection, excessive temperature, or abnormal heart-rate conditions, it identifies the situation as UNSAFE. This provides an early indication of hazardous working conditions and helps supervisors take appropriate safety measures.

4. System Architecture:

1. Worker Monitoring Layer

- The system is placed with or near the construction worker to continuously monitor health and environmental conditions.

2. Pulse Monitoring

- The Pulse Sensor measures the worker's pulse signal.
- The ESP8266 processes the signal to determine the worker's heart-rate condition.

3. Environmental Monitoring

- The DHT11 measures the surrounding temperature and humidity.
- The MQ2 sensor detects smoke and combustible gas conditions.

4. Location Monitoring

- The NEO-6M GPS module provides the worker's latitude, longitude, speed, and satellite information.
- This helps identify the worker's location during normal operation or unsafe conditions.

5. Processing Unit

- The ESP8266 NodeMCU acts as the central controller.
- It collects data from all connected sensors.
- It compares sensor values with predefined safety limits.

6. Safety Assessment

- The ESP8266 determines whether the worker's conditions are SAFE or UNSAFE.
- Unsafe conditions can include abnormal heart rate, high temperature, or gas detection.

7. Wireless Communication

- The ESP8266 uses Wi-Fi to transmit the collected sensor data to the cloud.

8. Cloud Monitoring

- ThingSpeak stores the sensor readings received from the ESP8266.
- The data can be displayed using numerical values, charts, and graphs.

9. Remote Monitoring

- Supervisors can remotely monitor worker health, environmental conditions, GPS location, and safety status through the ThingSpeak dashboard.

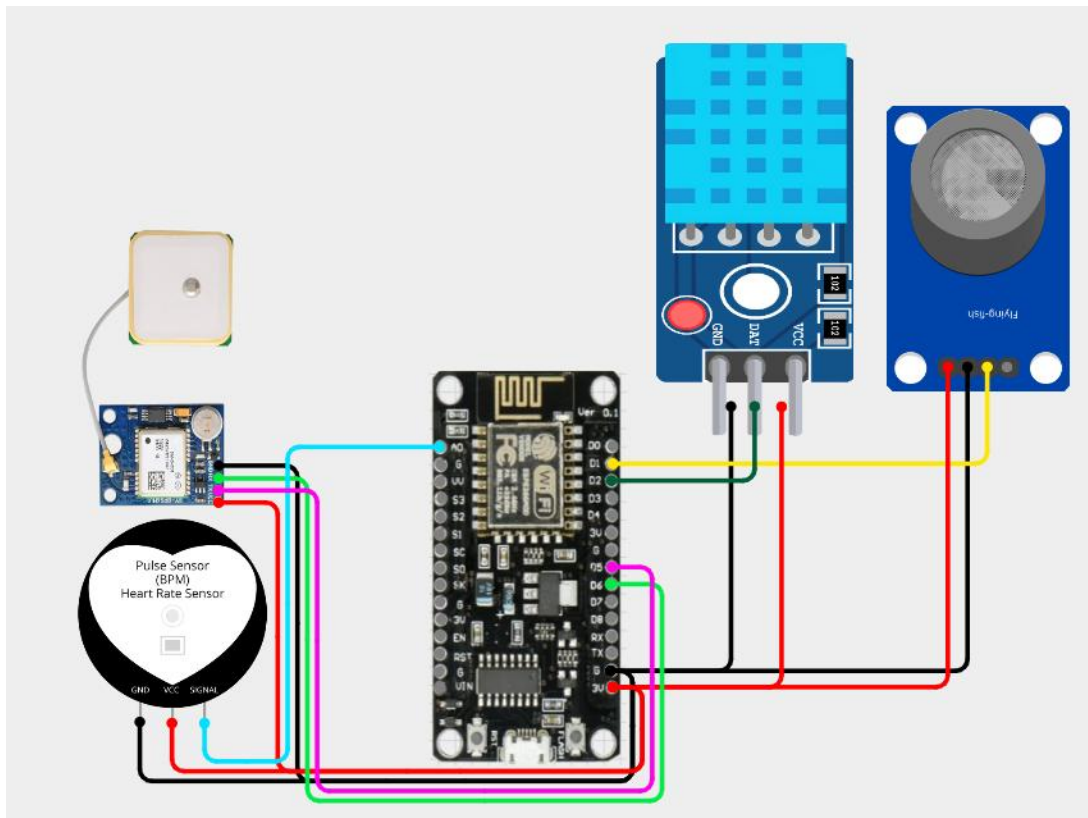
10. Data Analysis and Safety Response

- Historical sensor data can be analyzed to identify hazardous conditions and safety trends.
- The system supports faster decision-making and appropriate safety action when unsafe conditions are detected.

5.Circuit Table:

Component	Pin	ESP 32
Pulse Sensor	VCC	3.3V
Pulse Sensor	SIGNAL	A0
Pulse Sensor	GND	GND
DHT11	DATA	GPIO 4
DHT11	VCC	5V
DHT11	GND	GND
MQ2 Gas Sensor	D0	GPIO 5
MQ2 Gas Sensor	VCC	5V
MQ2 Gas Sensor	GND	GND
NEO-6M GPS	TX	GPIO 14
NEO-6M GPS	RX	GPIO 12
NEO-6M GPS	VCC	3.3V

6. Circuit Design:



7. Algorithm:

1. **Start and Initialize** – Power on the ESP8266 and initialize the Pulse Sensor, DHT11, MQ2, and NEO-6M GPS.
2. **Wi-Fi Connection** – Connect the ESP8266 to the configured Wi-Fi network for cloud communication.
3. **Collect Sensor Data** – Read the worker's heart rate, temperature, humidity, gas detection status, and GPS location.
4. **Process Data** – The ESP8266 processes the sensor readings and compares them with predefined safety limits.
5. **Safety Evaluation** – If abnormal heart rate, high temperature, or gas detection is identified, the system marks the condition as UNSAFE; otherwise, it marks it as SAFE.

6. GPS Monitoring – Obtain valid latitude, longitude, speed, and satellite information from the NEO-6M GPS module.

7. Cloud Upload – Send all collected parameters and the safety status to the corresponding ThingSpeak fields through Wi-Fi.

8. Display and Repeat – Display the current readings on the Serial Monitor and continuously repeat the monitoring and data-upload process.

8. Coding:

```
#include <ESP8266WiFi.h>
#include <ESP8266HTTPClient.h>
#include <WiFiClient.h>
#include <DHT.h>
#include <TinyGPSPlus.h>
#include <SoftwareSerial.h>

const char* WIFI_SSID = "SK2509";
const char* WIFI_PASSWORD = "SRDS2506";
const char* THINGSPEAK_API_KEY = "5U30FAOS0SXE1ET9";
const char* THINGSPEAK_SERVER = "http://api.thingspeak.com/update";
#define DHT_PIN 4
#define DHT_TYPE DHT11
DHT dht(DHT_PIN, DHT_TYPE);
#define PULSE_PIN A0
#define MQ2_PIN 5
#define GPS_RX_PIN 14
#define GPS_TX_PIN 12
SoftwareSerial gpsSerial(GPS_RX_PIN, GPS_TX_PIN);
TinyGPSPlus gps;
float temperature = 0.0;
float humidity = 0.0;
```

```
int pulseValue = 0;
int heartRate = 0;
int gasStatus = 0;
float latitude = 0.0;
float longitude = 0.0;
float gpsSpeed = 0.0;
bool gpsFix = false;
unsigned long lastSensorRead = 0;
unsigned long lastDisplay = 0;
unsigned long lastThingSpeak = 0;
const unsigned long SENSOR_INTERVAL = 1000;
const unsigned long DISPLAY_INTERVAL = 2000;
const float MAX_TEMPERATURE = 40.0;
const int MIN_HEART_RATE = 50;
const int MAX_HEART_RATE = 120;
void setup()
{
  Serial.begin(115200);
  delay(1000);
  Serial.println();
  Serial.println("=====");
  Serial.println(" WORKER HEALTH & SAFETY MONITORING SYSTEM");
  Serial.println("=====");
  pinMode(PULSE_PIN, INPUT);
  pinMode(MQ2_PIN, INPUT);
  gpsSerial.begin(9600);
  Serial.println();
  Serial.println("GPS CONFIGURATION");
  Serial.println("-----");
  Serial.println("NEO-6M TX -> D5 / GPIO14");
  Serial.println("NEO-6M RX -> D6 / GPIO12");
  Serial.println("GPS Baud -> 9600");
```

```
Serial.println();
Serial.println("GPS started...");
Serial.println("Waiting for GPS data...");
Serial.println();
Serial.print("Connecting to WiFi");
WiFi.begin(WIFI_SSID, WIFI_PASSWORD);
while (WiFi.status() != WL_CONNECTED)
{
    readGPS();
    delay(300);
    Serial.print(".");
}
Serial.println();
Serial.println("WiFi Connected!");
Serial.print("IP Address: ");
Serial.println(WiFi.localIP());
Serial.println();
Serial.println("SYSTEM READY");
Serial.println();
}

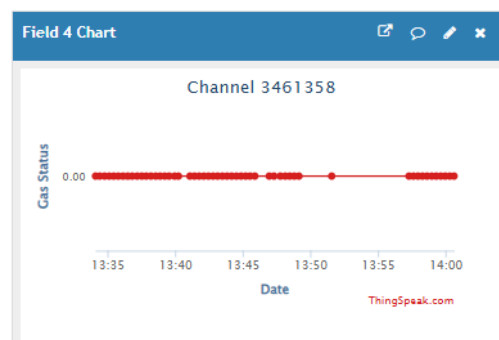
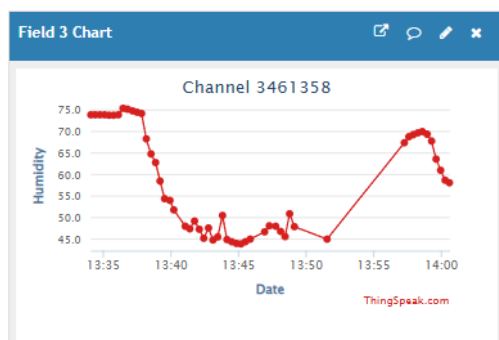
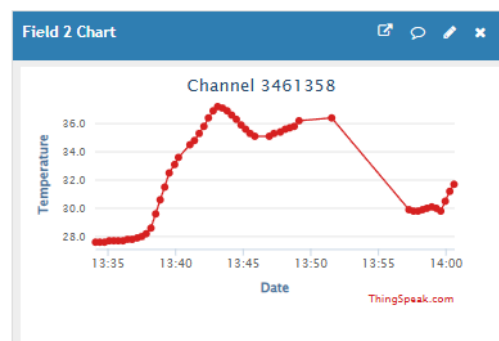
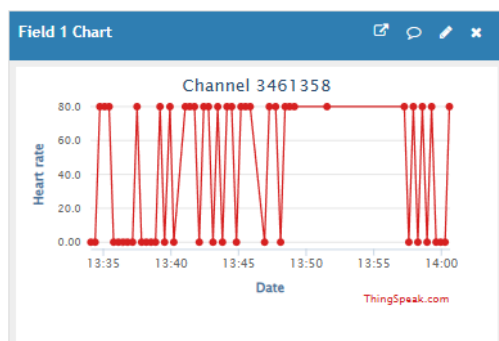
void readGPS()
{
    while (gpsSerial.available() > 0)
    {
        char c = gpsSerial.read();
        gps.encode(c);
    }
    if (gps.location.isValid())
    {
        latitude = gps.location.lat();
        longitude = gps.location.lng();
        gpsFix = true;
    }
}
```

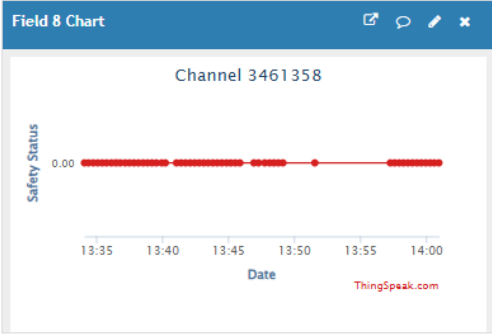
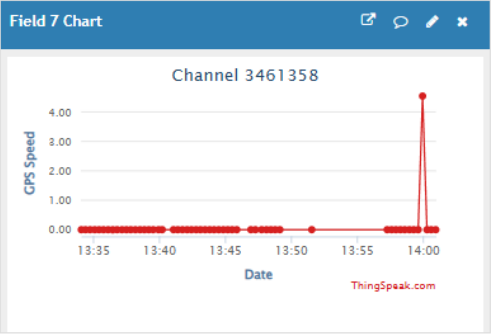
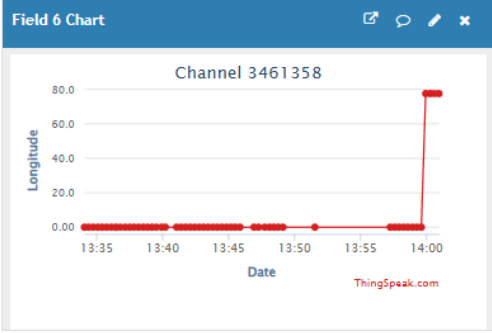
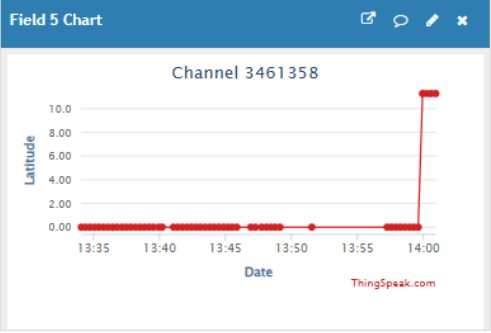
```
}  
else  
{  
    gpsFix = false;  
}  
if (gps.speed.isValid())  
{  
    gpsSpeed = gps.speed.kmph();  
}  
}  
void readDHT()  
{  
    float t = dht.readTemperature();  
    float h = dht.readHumidity();  
    if (!isnan(t))  
    {  
        temperature = t;  
    }  
    if (!isnan(h))  
    {  
        humidity = h;  
    }  
}  
void readMQ2()  
{  
    int state = digitalRead(MQ2_PIN);  
    if (state == LOW)  
    {  
        gasStatus = 1;  
    }  
    else  
    {
```

```
    gasStatus = 0;
}
}
void readPulse()
{
    pulseValue = analogRead(PULSE_PIN);
    if (pulseValue > 550)
    {
        heartRate = 80;
    }
    else
    {
        heartRate = 0;
    }
}
bool isUnsafe()
{
    if (gasStatus == 1)
    {
        return true;
    }
    if (temperature > MAX_TEMPERATURE)
    {
        return true;
    }
}
```

9. System Working:

1. The ESP8266 NodeMCU acts as the main controller and initializes all sensors and the Wi-Fi connection.
2. The Pulse Sensor monitors the worker's pulse and provides heart-rate information through the A0 pin.
3. The DHT11 sensor measures the surrounding temperature and humidity to identify uncomfortable or hazardous environmental conditions.
4. The MQ2 sensor, connected through its digital output to D1, detects smoke or combustible gases and provides a gas detection status.
5. The NEO-6M GPS module continuously receives satellite information and provides the worker's latitude, longitude, speed, and GPS status.
6. The ESP8266 processes all sensor readings and compares them with predefined safety limits.
7. If conditions such as high temperature, gas detection, or abnormal heart rate are detected, the system identifies the worker's condition as UNSAFE. Otherwise, it indicates SAFE.
8. The ESP8266 connects to the Internet through Wi-Fi and uploads the collected data to ThingSpeak Cloud.
9. ThingSpeak stores and displays the worker's health parameters, environmental conditions, GPS information, and safety status using numerical values and graphs.
10. The system continuously repeats this process, providing real-time remote monitoring of construction worker safety.





GPS INFORMATION

GPS Characters : 98271
GPS Sentences : 29
GPS FIX : YES
Latitude : 11.270604
Longitude : 77.603996
GPS Speed : 1.72 km/h
Satellites : 11
HDOP : 1.00
Altitude : 86.30 m
GPS Date : 19/08/2026
GPS Time : 08:30:07

SAFETY STATUS : SAFE

WORKER SAFETY DATA

Pulse ADC : 567
Heart Rate : 80 BPM
Temperature : 30.9 C
Humidity : 59.2 %
Gas Status : SAFE

GPS INFORMATION

GPS Characters : 99691
GPS Sentences : 32
GPS FIX : YES
Latitude : 11.270588
Longitude : 77.603981
GPS Speed : 1.80 km/h
Satellites : 10
HDOP : 1.10
Altitude : 88.50 m
GPS Date : 19/08/2026
GPS Time : 08:30:09

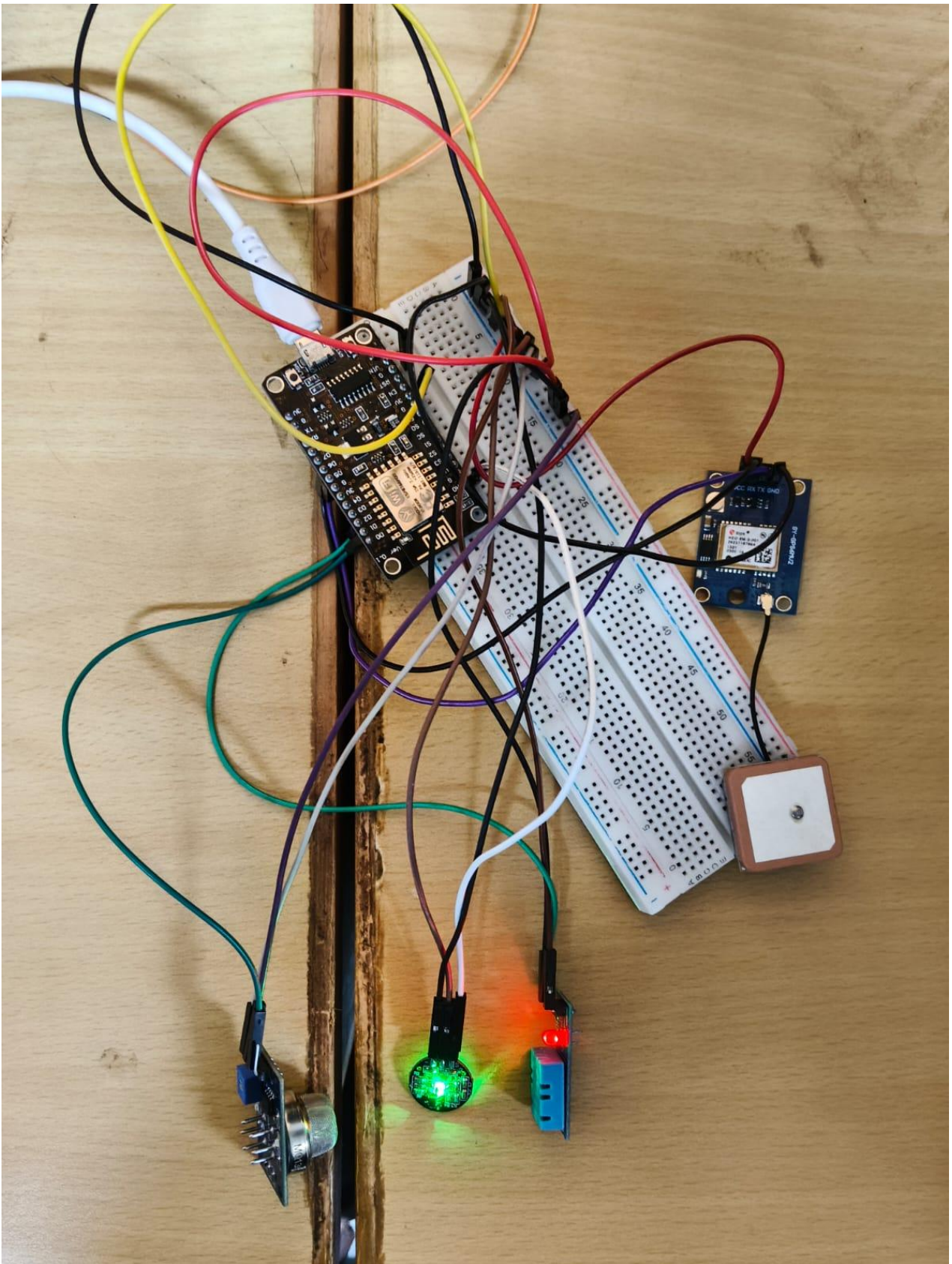
SAFETY STATUS : SAFE

10. Bill of Materials:

SNO	COMPONENT	QUANTITY
1	ESP8266 NodeMCU	1
2	Pulse Sensor	1
3	DHT11 Sensor	1
4	MQ2 Gas Sensor	1
5	NEO-6M GPS Module	1
6	Breadboard	1
7	USB Cable	1
8	ThingSpeak Account	1
9	Wi-Fi Network	1
10	Jumper Wires	Required

11. Prototype Implementation:

- 1. Hardware Assembly** – The ESP8266 NodeMCU is placed on the breadboard and connected to the Pulse Sensor, DHT11, MQ2 gas sensor, and NEO-6M GPS according to the circuit connections.
- 2. Sensor Integration** – The Pulse Sensor is connected to A0, MQ2 digital output to D1, DHT11 data to D2, and NEO-6M GPS TX/RX to D5/D6 respectively.
- 3. Programming** – The ESP8266 is programmed using Arduino IDE with libraries for DHT11, TinyGPSPlus, SoftwareSerial, Wi-Fi, and ThingSpeak communication.
- 4. Wi-Fi and Cloud Setup** – The ESP8266 is configured with the Wi-Fi credentials and ThingSpeak Write API Key. Eight ThingSpeak fields are created for heart rate, temperature, humidity, gas status, latitude, longitude, GPS speed, and safety status.
- 5. Sensor Data Collection** – The prototype continuously collects health, environmental, and location data from the connected sensors.
- 6. Safety Evaluation** – The ESP8266 compares the collected values with predefined limits and identifies the worker's condition as SAFE or UNSAFE.
- 7. GPS Tracking** – When the NEO-6M obtains a valid satellite fix, the worker's latitude, longitude, and speed are recorded.
- 8. Cloud Monitoring** – The processed data is transmitted to ThingSpeak through Wi-Fi at regular intervals, allowing the parameters to be monitored remotely.
- 9. Testing** – The prototype is tested under normal and unsafe conditions, such as increased temperature or gas detection, to verify that the system correctly updates the safety status.
- 10. Continuous Operation** – The system repeatedly monitors the worker and updates the cloud dashboard, providing continuous safety and location information



12.Results & Performance Analysis:

1. The developed prototype successfully monitors the worker's heart rate, temperature, humidity, gas status, and GPS location using the ESP8266.
2. The Pulse Sensor provides pulse information that can be used to identify abnormal heart-rate conditions.
3. The DHT11 successfully measures temperature and humidity, helping identify uncomfortable or potentially unsafe environmental conditions.
4. The MQ2 sensor detects smoke or gas through its digital output and provides a Safe/Gas Detected status.
5. The NEO-6M GPS provides the worker's latitude, longitude, speed, and satellite information when a valid GPS fix is available.
6. The ESP8266 successfully processes the sensor readings and determines the overall SAFE or UNSAFE condition based on predefined limits.
7. The sensor data is successfully transmitted to ThingSpeak Cloud through Wi-Fi and displayed using different fields and graphs.
8. The system provides continuous remote monitoring, allowing the collected worker safety parameters to be viewed from the ThingSpeak dashboard.
9. The prototype demonstrates that combining health monitoring, environmental sensing, GPS tracking, and cloud connectivity can provide an effective solution for construction worker safety.
10. Overall, the system shows good potential for real-time worker monitoring and early identification of unsafe working conditions, while further calibration and testing would be required for industrial deployment.